Personal information

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Professional experience

2026 - Present	Junior Professor Chair	Université de Strasbourg (France)
2025 - 2026	Assistant Professor (non-tenure track), Head of the Rock Physics and Geofluids Laboratory	École polytechnique fédérale de Lausanne (EPFL, Switzerland)
2022 - 2025	Research and Teaching Associate, Head of the Rock Physics and Geofluids Laboratory	École polytechnique fédérale de Lausanne (EPFL, Switzerland)
2021	Project Manager of the Interdisciplinary Thematic Institute for Geosciences for the Energy System Transition (ITI GeoT)	Université de Strasbourg (France)
2018 - 2020	Project Manager of the Laboratory of Excellence (LabEx) for Deep Geothermal Energy	Université de Strasbourg (France)
2016 - 2018	Post-doctoral researcher	Insitut du physique de Globe de Strasbourg (IPGS, France)
2013 - 2016	Doctoral student	Université d'Orléans (France)
2010 - 2012	Teaching and research assistant / Master student	University of British Columbia (Canada)
2008 - 2009	Consulting geophysicist	Golder Associates Ltd. (Canada)

Education

Date	Diplôme / Degree	Etablissement-Lieu / Institution-Location
03/2013 – 06/2016	Doctorat (PhD) en Science de l'Univers **Thesis title:** *Permeability development and evolution in volcanic systems: Insights from nature and laboratory experiments* (Dr. C. Martel, thesis advisor)	Université d'Orléans (France)
09/2010 – 09/2012	Master of Science (MSc), Geological Sciences **Thesis title:** *An experimental investigation of the mechanical behaviour of synthetic calcite-dolomite composites* (Dr. L.A. Kennedy and Prof. J.K. Russell, thesis advisors)	University of British Columbia (Canada)
09/2003 – 05/2008	Bachelor of Science (BSc), Geophysics honours program **Thesis title:** *Understanding Multi-Peak Anomalies for Unexploded Ordinance Discrimination* (Prof. D. Oldenburg, thesis advisor)	University of British Columbia (Canada)

Scientific expertise

I use experimental techniques including rock and magma deformation and rock-fluid interaction experiments under *in situ* pressure and temperature conditions to understand the structure and behaviour of geomaterials. Most of my research has focused on the evolution of permeability in geosystems, with particular emphasis on volcanic hazards and geothermal energy exploitation. I am adept at rock physical property characterization (e.g. porosity, permeability, rock strength, thermal conductivity, seismic properties) and use analytical techniques to quantify the microstructure and chemistry of rocks (using optical microscope, SEM, EBSD, XRPD, XCT, and EMPA). From 2022 to 2026, I was head of the Rock Physics and Geofluids Laboratory at EPFL whose mission is the study of how rock physical properties evolve over time in response to rock-fluid interactions under volcanic and geothermal conditions. In my capacity as unit head, I was responsible for the establishment and advancement of our scientific program, direct supervision of my PhD students, as well as the management of my unit. I am now Junior Professor Chair in Experimental Rock Physics (Rock-Fluid Interactions) at the University of Strasbourg, France.

Keywords

Rock-fluid interactions, Experimental rock physics, High temperature / high pressure experiments, Volcanology, Geothermal energy

Major events in scientific career

Research: In 2022, I established the Rock Physics and Geofluids Laboratory (RPGL) at EPFL where my research group and I study the role that rock-fluid interactions play in modifying the physical structure – and therefore the hydraulic and mechanical behaviours – of geomaterials.

Teaching: I have taught at the undergraduate and Master level at six different universities, including a Master course in Geothermal Resource Development (EPFL, 2025-2026). I am currently responsible for teaching at the Master Level at the Ecole et Observatoire des Sciences de la Terre (EOST) at the University of Strasbourg in Borehole geophysics, Introduction to the Energy Transition, and Geotechnics. I currently supervise two PhD students.

Competitive funding: I have been awarded over 3,400,000 € in individual funding, including two post-graduate fellowships from the Natural Sciences and Engineering Research Council of Canada (NSERC, 2011 and 2013) and more recently a Swiss National Research Foundation (SNSF) PRIMA Grant (2022 – 2026) and an SNSF Strating Grant (2025 – 2030).

Awards: I was awarded the **2023 Outstanding Early Career Scientist Award** by the European Geosciences Unions (EGU) Division of Earth Magnetism and Rock Physics (EMRP). In 2012, I was awarded a **Teaching Assistant Award** at the Department of Earth, Oceans, and Atmospheric Sciences (University of British Columbia, Canada) *"in recognition of [my] initiative and exceptional effort under unusual circumstances to take on responsibilities of organizing practicals and supporting the teaching of [Structural Geology II], going beyond what is expected of a teaching assistant."*

Keynote talks and invited seminars: I have given keynote talks at the 2023 European Geosciences Union General Assembly and 2023 EuroConference in Rock Physics and Rock Mechanics and have been invited to give seminars in nine countries.

Teaching experience

Teaching responsibilities

From 2026 **École et Observatoire des Sciences de la Terre, Université de Strasbourg (Strasbourg, France), Junior Professor Chair.** Courses include:

- Introduction to energy transition methods (Master, 24h/year)
- Geotechnics (Engineering school, 12h/year)
- Borehole geophysics (Engineering school, 18h/year)

2022-2025 **Institute of Civil Engineering, École polytechnique fédérale de Lausanne (EPFL, Switzerland), Lecturer.** Courses include:

- **Geothermal Resource Development** (Master, 42h/year, starting autumn 2024)
- **Rock Mass Characterisation** (Master, 5h/year)

2021 **French-Azerbaijani University (online, Azerbaijan), Guest lecturer.** Lectures include:

- **Renewable georesources: Geothermal energy** (Master, 6h)

2014-2021 **University of Strasbourg (France), Guest lecturer**. Lectures include:

- **Introduction to volcanic processes** (Master, 12h, 2016-2021) in the course "Rock physics applied to reservoirs and hazards"
- **Stress and Strain** (Master, 2h, 2018) and **Deformation Mechanisms and Microstructure** (2h, 2014) in the course "Brittle microstructure"
- **Reservoir permeability** (Master, 1h, 2018) in the course "Geothermal energy"
- **Petrophysics Practical** (Master, 6h, 2017)

2018 **Northeastern University (China), Guest lecturer:**

- **Scientific writing** (Undergraduate and graduate students, 4h)

2016 **University of Orléans (France), Guest lecturer:**

- **Introduction to experimentation** (Undergraduate, 4h)

2010-2012 **University of British Columbia (Canada), Substitute lecturer and teaching assistant** for undergraduate courses (Undergraduate; average of 6 hours per week). Courses include: Structural Geology I (3rd year), Structural Geology II (4th year), The Solid Earth: A Dynamic Planet (1st year), Earth's Treasures: Gold and Gems (1st year), The Earth and the Solar System (3rd year, Faculty of Arts).

International visibility

Awards and scholarships

2023 2023 EGU Division Outstanding Early Career Scientist Award – Division of Earth Magnetism and Rock Physics (EMRP)

2017 Early Career Grant, 12th Euro-Conference on Rock Physics and Geomechanics

2015 Outstanding Student Poster (OSP) Award, European Geosciences Union (EGU)

2012 Departmental Teaching Assistant Award, Department of Earth, Ocean, and Atmospheric Sciences for 2011-2012, University of British Columbia $500

2012 Scholarship of the Embassy of France 5.5k€

2012 Keith Runcorn Travel Award for Non-Europeans, European Geosciences Union (EGU) 440€

2010 University of British Columbia George E Winkler Memorial Scholarship $3k

2010 University of British Columbia Thomas and Marguerite MacKay Memorial Scholarship $6k

2008 Association of Professional Engineers and Geoscientists Achievement Award in Geoscience

2007 British Columbia Geophysical Society Scholarship $750

2007 University of British Columbia Thomas and Marguerite MacKay Memorial Scholarship $4k

2006 University of British Columbia Thomas and Marguerite MacKay Memorial Scholarship $1.2k

Keynote addresses at international conferences

2023: **Kushnir A.,** Heap, M.J., Baud, P., Reuschlé, T., Schmittbuhl, J. **To break or not to break: The role of secondary mineralisation on joint reactivation in sandstone.** Presented at 2023 Euro-Conference on Rock Physics and Rock Mechanics, Ziest, Netherlands, 23-26 October.

2023: **Kushnir A.,** Heap, M.J., Baud, P., Reuschlé, T., Schmittbuhl, J. **Reactivating sealed joints: rock strength reduction and permeability enhancement…sometimes.** Presented at European Geosciences Union (EGU) 2023 General Assembly, Vienna, Austria, 23-28 April.

2023: **Kushnir A. Permeability, alteration, and microstructure: A (hopefully) coupled rock physics and geochemical approach to how rock-fluid interactions change permeability.** Presented at European Geosciences Union (EGU) 2023 General Assembly, Vienna, Austria, 23-28 April.

Invited seminars

2026 **The trials, tribulations, and successes of using experimental approaches to understand how rock-sluid interactions change the permeability of rocks.** (2026) Insitut thématique interdisciplinaires Géoscience pour la transition énérgétique (ITI GeoT), École et Observatoire des Sciences de la Terre (EOST), Université de Strasbourg (Unistra), Invited by Caroline Correira, 2 March 2026, Strasbourg, France.

2025 **Permeability in the lab: Capabilities, advantages, and challenges.** (2025) Invited seminar as part of the *Permeability across scales Round Table*, University of Lausanne (UniL), Invited by Lucas Pimienta, 26 March 2025, Lausanne, Switzerland.

2024 **Alteration and permeability evolution in volcanic rocks.** (2024) Geociencias Barcelona (GEO3BCN-CSIC), Invited by Adelina Geyer Traver, 22 May 2024, Barcelona, Spain.

2023 **From the laboratory to the field: Making sense of rock physical property measurements at the borehole scale.** (2023) Geological Survey of Finland (GTK), Invited by Alan Bischoff, 3 May 2023, Helsinki, Finland.

2023 **ReSiDue: Redistribution of silica and deposition under a volcanic edifice.** (2023) Universidad de Chile, Invited by Luis Felipe Orellana, 5 January 2023, Santiago, Chile.

2022 **The Empty Space: How rock-fluid interactions modify permeability in volcanic systems.** (2022) University of Lausanne (UniL), Invited by Lukas Baumgartner, 5 December 2022, Lausanne, Switzerland.

2022 **The Empty Space: How rock-fluid interactions modify permeability in volcanic systems.** (2022) Institut national de la recherche scientifique (INRS), Invited by Jasmin Raymond, 18 July 2022, Québec, Canada.

2021 **The Empty Space: How rock-fluid interactions modify permeability in volcanic systems.** (2021) University of Bern, Invited by Joerg Hermann, 29 November 2021, Bern, Switzerland.

2020 Volcanica: Volcanology's first Diamond Open Access journal – Success and challenges. (2020) Invited speaker at the *University of Strasbourg's Open Access Week*, University of Strasbourg, Invited by Katherine Sowley, 5 November 2020, Strasbourg, France.

2020 Deep geothermal energy: A meeting place for rock physics and the energy systems transition. (2020) Invited seminar as part of the *Rock Mechacnics and the Energy Transition Day,* Comité Français de Mécanique des Roches (French Committee for Rock Mechanics, CFMR), Invited by Lina María Guayacan, 28 May 2020, Online.

2019 Volcanica: A community-based initiative pioneering diamond Open Access publishing in volcanology. (2019) Invited keynote address at the *2019 Doctoral School Congress*, University of Strasbourg, Invited by Lucille Carbillet, 27 November 2019, Strasbourg, France.

2018 Estimating down-borehole permeability in a geothermal reservoir using laboratory measurements. (2018) Northeastern University, Invited by Tao Xu, 23 October 2018, Shenyang, China.

2017 Breaking through: How magma becomes permeable in the laboratory. (2017) Canterbury University, Invited by Ben Kennedy, 10 August 2017, Christchurch, New Zealand.

2017 Let 'er rip: Why a permeable volcano is a happy volcano. (2017) University of Waikato, Invited by Shaun Barker, 1 August 2017, Hamilton, New Zealand.

2017 Breaking through: How magma becomes permeable in the laboratory. (2017) University of Savoie Mont Blanc, Invited by Marielle Collombet, 30 June 2017, Chambéry, France.

2016 Breaking through: How magma becomes permeable in the laboratory. (2016) Ludwig Maximilian University of Munich, Invited by Fabian Wadsworth, 24 November 2016, Munich, Germany.

2012 Carbonates in thrust faults: The role of coarse-grained dolomite in fine-grained limestone. (2012) School and Observatory of Earth Sciences, Invited by Renaud Toussaint, 25 October 2012, Strasbourg, France.

Collaboration in international projects

2026-Present BRITTLENESS : How does fluid chemistry affect the brittle-ductile transition in limestones; PI: Prof. Anne Pluymakers; Funded by the NWO Talent Programme Vidi Scheme, Netherlands

Reviewer activities

Participation in evaluation committees

Candidacy exams

2026 Micaela Castellani, Title to be announced. 11 June 2026, Institut du Physique du Globe de Paris, Paris, France.

2025 Mathilde Metral, Hydro-mechanical and gas transport behaviour of clay-based barriers in deep geological repositories. 23 September 2025, École polytéchnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

2024 Sophie Ten Bosch, Multiphysical modelling of sustainable geotechnics with a focus on biocementation and energy geostructures. 16 April 2024, École polytéchnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Master project defences

2025 Corentin Jaire, Upscaling of reactive transport for carbon capture and storage. 7 March 2025, École polytéchnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

2023 Cesare Griner, Hydromechanical impact of CO2 mineralisation in basaltic rock. 1 July 2023, École polytéchnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.

Participation in PhD defence committees

2026 **François Décossin,** Modélisation thermocinétique expérimentale et numérique de l'altération hydrothermale des roches volcaniques - Exemple de la Soufrière de Guadeloupe (Caraïbes orientales, France). 26 June 2026, Université d'Orléans, Orléans, France.

2025 **Theo Briolet,** Effect of alteration on the hydromechanical properties of rocks. 24 March 2025, Université PSL, ENS-Paris, and IFPEN, Paris, France.

2024 **Milad Naderloo,** Fault reactivation and rock deformation mechanisms under stress/pressure cycling. 13 September 2024, TU Delft, Delft, Netherlands.

2024 **Héloïse Fuselier,** Thermo-hydro-mechanical behaviour of the Callovo-Oxfordian (COx) claystone under thermal changes. 21 March 2024, EPFL, Lausanne, Switzerland.

Review of articles or projects

2026-Present Associate Editor of AGU's Journal of Geophysical Research: Solid Earth

2017-Present Founding member, editor, editorial committee member (from 2017 to 2024) at Volcanica, the first and only Diamond Open Access peer-reviewed scientific journal in volcanology

2018-2020 Guest editor of the Springer Geothermal Energy Special Issue "From Exploration to Operation – Research Developments in Deep Geothermal Energy"

Reviewer for major international journals: Nature Communications, Journal of Geophysical Research: Solid Earth, Nature Scientific Reports, Journal of Volcanology and Geothermal Research, Geothermics, Journal of Petroleum Science and Engineering, Chemical Geology, Volcanica, Bulletin of Volcanology, Heliyon, Solid Earth, Environment Earth Sciences, Advances in Water Research,

Service to the community

Leading teams, services, research and/or training departments

2022 – 2026 Director of the Rock Physics and Geofluids Laboratory, Institute of Civil Engineering, École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland

Participation in the life of the research unit or institution

2022-2025 Member of the ENAC Diversity Office at the École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

2022-2025 Member of the GEOscience Diversity & Equality Switzerland (GEODES) committee, Switzerland

2022-2024 Member of the Executive Committee of CLIMACT: Center for Climate Impact and Action at the Université de Lausanne (UNIL) and École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

International conference organisation

2023-2026 16th Euroconference on Rock Physics and Rock Mechanics, Local organizing committee, Les Diablerets, Switzerland, January 2026.

2025 IAVCEI Scientific Assembly 2025, Local organizing committee, Convener of "Hydrothermal alteration in volcanic systems" session, Geneva, Switzerland, July 2025.

2024 European Geosciences Union (EGU) General Assembly, Convener of "Hydrothermal alteration in volcanic systems" session, Vienna, Austria, April 2024.

2023 European Geosciences Union (EGU) General Assembly, Co-convener of "Volcanic hydrothermal alteration and fluid-rock interactions" session, Vienna, Austria, April 2023.

2018, 2020 European Geothermal Workshop, Local organizing committee, Strasbourg, France (October 2018) and Online (2020).

Popularizing science

Videos

2020-2021 I supervised the creation of eight animated videos designed to communicate the different scientific missions of the Interdisciplinary Thematic Institute for Geosciences for the Energy System Transition (ITI GeoT) to the general public. This work was undertaken as a result of a Master project conducted by Jérôme Arnaud (see Encadrement / Supervision). All videos can be found here (I am the narrator of the English videos): https://geot.unistra.fr/science-society/scientific-mediation

Blog posts

2017 **Kushnir, A.R.L.** (2017) **A little fracture can go a long way: How experiments illuminate our understanding of volcanic eruptions.** EGU Geochemistry, Mineralogy, Petrology, and Volcanology Division Blog. 1 December 2017**.** https://blogs.egu.eu/divisions/gmpv/2017/12/01/a-little-fracture-can-go-a-long-way/

2017 **Kushnir A.R.L.** (2017) **Into the belly of the beast.** EGU blog: GeoLog. 17 April 2017. https://blogs.egu.eu/geolog/2017/04/17/imaggeo-on-mondays-in-the-belly-of-the-beast/

Research funding

Individual grants (Total: > 3,450,000 €)

Year	Source (Funding body)	Grant program	Project title	Coordinator names	Budget (€)	Role in project
2025 – 2030 (awarded in 2024)	Swiss National Science Foundation (SNSF)	SNSF Starting Grant	FuSE-EGS: Fracture sealing and reactivation in Enhanced Geothermal Systems	Alexandra Kushnir	1,900,000 € (1,800,000 CHF)	Principal investigator
2022 – 2026 (awarded in 2021)	Swiss National Science Foundation (SNSF)	PRIMA Grant	ReSiDue: Redistribution of silica and deposition under a volcanic edifice	Alexandra Kushnir	1,500,000 € (1,400,000 CHF)	Principal investigator
2013 - 2016	National Science and Engineering Research Council of Canada (NSERC)	Canadian Postgraduate Scholarship (PGS D)	Effusive-explosive transitions in silicic eruptions: Experimental and field-based investigations	Alexandra Kushnir	40,000 € ($63,000 Cdn)	Principal investigator
2011	National Science and Engineering Research Council of Canada (NSERC)	Alexander Graham Bell Canada Graduate Scholarship (CGS M)	An experimental investigation of the mechanical behavior of synthetic calcite-dolomite composites	Alexandra Kushnir	11,000 € ($17,500 Cdn)	Principal investigator

Initiative grants (Total: > 200,000 €)

Year	Source (Funding body)	Grant program	Project title	Coordinator names	Budget (€)	Role in project
2025	Swiss National Science Foundation (SNSF)	Scientific exchanges	16th Euroconference for Rock Physics and Rock Mechanics	Marie Violay Alexandra Kushnir	26,000 € (25,000 CHF)	Co-principal investigator
2023	École polytechnique fédérale de Lausanne (EPFL)	2023 ENAC Call for Equipment	High temperature furnace	Alexandra Kushnir Marie Violay	18,000 € (17,000 CHF)	Co-principal investigator
2022	École polytechnique fédérale de Lausanne (EPFL)	2022 ENAC Call for Equipment	High temperature / high pressure reactor cells	Alexandra Kushnir Marie Violay	53,000 € (50,000 CHF)	Co-principal investigator
2022	École polytechnique fédérale de Lausanne (EPFL)	2022 ENAC Call for Equipment	High pressure permeameter	Marie Violay Alexandra Kushnir	53,000 € (50,000 CHF)	Co-principal investigator
2022	Initiative d'excellence (IdEx) de l'Université de Strasbourg	University and city: "Open Science" program	Cofunding for: Volcanica: Growing an innovative community open access testbed in the Earth Sciences,	Michael Heap Alexandra Kushnir	11,000 €	Co-principal investigator
2021	French National Fund for Open Science (FNSO)	Second call for projects: "Publications"	Volcanica: Growing and innovative community open access testbed in the Earth Sciences	Michael Heap Alexandra Kushnir Jamie Farquharson Fabian Wadsworth	45,000 €	Co-principal investigator

Supervision

Doctoral student supervision

Name & Surname	Cindy Mikaelian	In progress	In progress Defence expected December 2026	Level of supervision (0 to 100)	95%
Institution		École Polytechnique Fédérale de Lausanne			
Thesis title		***Redistribution of silica and deposition under a volcanic edifice: A geochemical and rock physics approach***			
Current situation		Thesis in progress			

Name & Surname	Andrea Mazzeo	In progress	In progress Defence expected March 2027	Level of supervision (0 to 100)	95%
Institution		École Polytechnique Fédérale de Lausanne			
Thesis title		***An experimental rock physics and analytical modelling approach to permeability modification by rock alteration***			
Current situation		Thesis in progress			

Master student supervision

Name & Surname of student	Supervisor role	Degree programme	Year
Mathilde Janottin	Supervisor	Géotechnique, Génie civil, Ecole polytechnique fédérale de Lausanne : ***Effect of fluid chemistry on the mechanical behaviour of basalt with application to chemical stimulation in geothermal reservoirs***	2025
Jérôme Arnaud	Supervisor	Master de Communication Scientifique, Université de Strasbourg: ***Creating audio-visual support materials for the ITI Geosciences for the Energy System Transition.***	2020
Pauline Harlé	Co-supervisor (Main supervisor: P; Duringer)	Master sciences de la Terre et des planets, environnement, GDT, Université de Strasbourg: ***Nature and conductivity of subsurface sedimentary rocks: application to the assessment of the geothermal gradient in the Upper Rhine Graben***	2018

Other supervision

Name & Surname of student	Supervisor role	Degree programme	Year
Carla Regenass and Aurélien Sapin	Primary supervisor	*Bachelor semester project, Génie Civil, École polytechnique fédérale de Lausanne*: ***The role of secondary mineralisation on the hydraulic properties of fractures in geothermal systems***	2025
Sebastiano Giovanettina	Co-supervisor (Co-supervisor: M. Violay)	*Bachelor semester project, Génie Civil, École polytechnique fédérale de Lausanne:* ***Calibration and benchmarking of new gas permeameter***	2023

Julien Schneider	Co-supervisor (Main supervisor : P. Baud)	*2nd year engineering research internship, EOST, University of Strasbourg:* ***Thermal conductivity of rocks from Soultz-sous-Forêts using the Hot Disk method***	2018

Scientific works

Summary of scientific works

Articles in peer-review scientific journals	37
Peer-reviewed conference proceedings	5
First author oral contributions to conferences	11
First author poster contributions to conferences	5
Co-authored contributions to conferences (oral and posters)	27
Editorials	2
Technical / government reports	3

Articles in peer-review scientific journals

h-index: 24; i10-index: 32; Citations: >1700
37 articles published in peer-reviewed scientific journals; citation data from Google Scholar, March 2026

1. Mendieta, A., Rosas-Carbajal, M., Jessop, D.E., Komorowski, J-C., Deroussi, S., Vlastelic, I., Villeneuve, M., Violay, M., Ongaro, T.E., **Kushnir, A.R.L.**, Beauducel, F., de Chabalier, J-B., Bajou, R., Baud, P., Heap, M.J. (2026) Geophysics-assisted imaging of shallow gas pockets and hydrothermal alteration at the summit of La Soufrière de Guadeloupe (Eastern Caribbean). ***Bulletin of Volcanology.*** *In press.*

2. Heap, M.J., Mitchell, S.J., Carey, R.J., **Kushnir, A.R.L.** (2025) Influence of cristobalite and diktytaxitic textures on the physical properties of submarine rhyolite lavas from Havre volcano (Kermadec volcanic arc, Pacific Ocean). ***Journal of Volcanology and Geothermal Research.*** 108392. DOI: https://doi.org/10.1016/j.jvolgeores.2025.108392. *Citations: -.*

3. Xu, B., Xu, T., Du, S., Heap, M.J., **Kushnir, A.R.L.**, Liu, B. (2025) Time-dependent deformation of sandstone due to chemical corrosion: An investigation using the heterogeneous grain-based phase-field method. ***Rock Mechanics and Rock Engineering.*** 1-27. DOI: https://doi.org/10.1007/s00603-025-04439-2. *Citations: 6.*

4. Xu, B., Xu, T., Heap, M.J., **Kushnir, A.R.L.**, Su, B., Lan, X. (2024) An adaptive phase field approach to 3D internal crack growth in rocks. ***Computers and Geotechnics.*** 173: 106551. DOI: https://doi.org/10.1016/j.compgeo.2024.106551. *Citations: 26.*

5. Heap, M.J., Bayramov, K., Meyer, G.G., Violay, M.E.S., Reuschlé, T., Baud, P., Gilg, H.A., Harnett, C.E., **Kushnir, A.R.L.**, Lazari, F., Mortensen, A.K. (2024) Compaction and permeability evolution of tuffs from Krafla Volcano (Iceland). ***Journal of Geophysical: Solid Earth.*** 8: e2024JB029067. DOI: https://doi.org/10.1029/2024JB029067. *Citations: 12.*

6. **Kushnir, A.R.L.,** Heap, M.J., Baud, P., Reuschlé, T., Schmittbuhl, J. (2023) Reactivation of variably-sealed joints and permeability enhancement in geothermal reservoir rocks. ***Geothermal Energy.*** 11(23). DOI: https://doi.org/10.1186/s40517-023-00271-5. *Citations: 9.*

7. Caselle, C., Baud, P., **Kushnir, A.R.L.**, Reuschlé, T., Bonetto, S.M.R. (2022) Influence of water on deformation and failure of gypsum rock. ***Journal of Structural Geology.*** 104722. DOI: https://doi.org/10.1016/j.jsg.2022.104722. *Citations: 33.*

8. Heap, M.J., Jessop, D.E., Wadsworth, F.B., Rosas-Carbajal, M., Komorowski, J-C., Gilg, H.A., Aron, N., Buscetti, M., Gential, L., Goupil, M., Masson, M., Hervieu, L., **Kushnir, A.R.L.**, Baud, P., Carbillet, L., Ryan, A.G., Moretti, R. (2022) The thermal properties of hydrothermally altered andesites from La Soufrière de Guadeloupe (Eastern Caribbean). ***Journal of Volcanology and Geothermal Research.*** 421, 107444. DOI : https://doi.org/10.1016/j.jvolgeores.2021.107444. *Citations: 27.*

9. Wadsworth, F.B., Vossen, C.E.J., Heap, M.J., **Kushnir, A.**, Farquharson, J.I., Schmid, D., Dingwell, D.B., Belohlavek, L., Huebsch, M., Carbillet, L., Kendrick, J.E. (2021) The force required to operate the plunger on a French press. ***American Journal of Physics.*** 89, 769. DOI: https://doi.org/10.1119/10.0004224. *Citations: 12.*

10. Martel, C., Pichavant, M., Di Carlo, I., Champallier, R., Wille, G., Castro, J.M., Devineau, K., Davydova, V.O., **Kushnir, A.R.L.** (2021) Experimental constraints on the crystallization of silica phases in silicic magmas. ***Journal of Petrology.*** egab004. DOI: https://doi.org/10.1093/petrology/egab004. *Citations: 21.*

11. **Kushnir, A.R.L.**, Heap, M.J., Griffiths, L., Wadsworth, F.B., Langella, A., Baud, P., Reuschlé, T., Kendrick, J.E. and Utley, J.E. (2021) The fire resistance of high-strength concrete containing natural zeolites. ***Cement and Concrete Composites***. 116, 103897. DOI: https://doi.org/10.1016/j.cemconcomp.2020.103897. *Citations: 44.*

12. Wadsworth, F.B., Vasseur, J., Llewellin, E.W., Brown, R.J., Tuffen, H., Gardner, J.E., Kendrick, J.E., Lavallée, Y., Dobson, K.J., Heap, M.J., Dingwell, D.B., Hess, K-U., Schauroth, J., von Aulock, F.W., **Kushnir, A.R.L.**, Marone F. (2020) A model for permeability evolution during volcanic welding. ***Journal of Volcanology and Geothermal Research.*** 107118. DOI: https://doi.org/10.1016/j.jvolgeores.2020.107118. *Citations: 34.*

13. Farquharson, J.I.**, Kushnir, A.R.L.**, Wild, B., Baud, P. (2020) Physical property evolution of granite during experimental chemical stimulation. ***Geothermal Energy***. 8(1): 14. DOI: https://doi.org/10.1186/s40517-020-00168-7. *Citations: 31.*

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First author oral contributions to conferences

1. **Kushnir, A.R.L.,** Heap, M.J., Baud, P., Reuschlé, T., Schmittbuhl, J. (2023) To break or not to break: The role of secondary mineralisation on joint reactivation in sandstone. Presented at 2023 Euro-Conference on Rock Physics and Rock Mechanics, 23-26 October 2023, Zeist, Netherlands. **(Keynote)**
2. **Kushnir, A.R.L.,** Heap, M.J., Baud, P., Reuschlé, T., Schmittbuhl, J. (2023) Reactivating sealed joints: rock strength reduction and permeability enhancement…sometimes. Presented at European Geosciences Union (EGU) 2023 General Assembly, 23-28 April 2023, Vienna, Austria. **(Solicited)**
3. **Kushnir, A.R.L.** (2023) Permeability, alteration, and microstructure: A (hopefully) coupled rock physics and geochemical approach to how rock-fluid interactions change permeability. Presented at European Geosciences Union (EGU) 2023 General Assembly, 23-28 April 2023, Vienna, Austria. **(Keynote)**
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5. **Kushnir, A.R.L.,** Heap, M.J., Baud, P., (2017) A journey through scales: Informing reservoir-scale fracture-controlled permeability using laboratory measurements. Presented at 12th Euro Conference on Rock Physics and Geomechanics, 5-10 November 2017, Jerusalem, Israel.
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11. **Kushnir, A.**, Kennedy, L.A., Misra, S., Benson, P., (2012). Carbonates in thrust faults: High temperature investigations into deformation processes in calcite-dolomite systems. Presented at European Geosciences Union (EGU) 2012 General Assembly, 22-27 April 2012, Vienna, Austria.

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1. **Kushnir, A. R. L.,** Mikaelian, C., Mazzeo, A. (2024) Introducing ReSiDue: A coupled rock physics and geochemical approach to understanding how rock-fluid interactions change permeability in volcanic systems. Presented at European Geosciences Union (EGU) 2024 General Assembly, 14-19 April 2024, Vienna, Austria.
2. **Kushnir, A. R. L.,** Mikaelian, C., Mazzeo, A. (2024) ReSiDue: A coupled rock physics and geochemical approach to understanding how rock-fluid interactions change permeability in volcanic systems. Presented at 11th Minisymposium on Poroelasticity and Rock Physics 2024, 5 March 2024, Lausanne, Switzerland.
3. **Kushnir, AR.L.,** Heap, M.J., Baud, P. (2018) Thermal properties of the sedimentary cover (Muschelkalk and Buntsandstein) at the Soultz-sous-Forêts (France) geothermal site. EGU2018-16574. Presented at European Geosciences Union (EGU) 2018 General Assembly, 8-13 April 2018, Vienna, Austria.

4. **Kushnir, A.R.L.**, Schneider, J., Heap, M.J., Lengliné, O., Baud, P., (2018) Thermal properties of the Muschelkalk and Buntsandstein from the EPS-1 borehole at Soultz-sous-Forêts (France). Presented at 6th European Geothermal Workshop, 10-11 October 2018, Strasbourg, France.

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Co-authored contributions to conferences (oral and posters)

1. Mikaelian, C., Baumgartner, L., Adams, A., Violay, M., **Kushnir, A.R.L.** Cristobalite formation in synthetic glasses through the role of mineralizing agents in alteration experiments. Presented at 2025 IAVCEI Scientific Assembly, 29 June-4 July 2025, Geneva, Switzerland.

2. Mazzeo, A., Violay, M., **Kushnir, A.R.L.** Reactive flow-induced alteration and permeability evolution in volcanic rock analogues. Presented at 2025 IAVCEI Scientific Assembly, 29 June-4 July 2025, Geneva, Switzerland.

3. Mendieta, A., Rosas-Carbajal, M., Jesspo, D., Bajou, R., Baud, P., Deroussi, S., Esposito Ongaro, T., Komorowski, J-C., **Kushnir, A.R.L.**, Villeneuve, M., Heap, M.J. Hydrothermal alteration and multi-phase 3D structure of the summit of La Soufrière de Guadeloupe. Presented at 2025 IAVCEI Scientific Assembly, 29 June-4 July 2025, Geneva, Switzerland.

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5. Heap, M. J., Bayramov, K., Meyer, G., Violay, M., Reuschlé, T., Baud, P., Gilg H. A., Harnett, C., **Kushnir, A.,** Lazari, F., Mortensen, A. Compaction and permeability evolution of tuffs from the Krafla geothermal system (Iceland). Presented at European Geosciences Union (EGU) 2025 General Assembly, 27 April-2 May 2025, Vienna, Austria.

6. Mendieta, A., Rosas-Carbajal, M., Bajou, R., Baud, P., Deroussi, S., Esposti Ongaro, T., Komorowski, J.-C., **Kushnir, A. R. L.,** Villeneuve, M., and Heap, M. J. Imaging hydrothermal alteration with electrical resistivity tomography in La Soufrière de Guadeloupe volcano. Presented at European General Assembly (EGU) 2024 General Assembly, 14–19 April 2024, Vienna, Austria.

7. Wadsworth, F.B., Llewellin, E.W., Vasseur, J., Tuffe, H., Foster, A., Unwin, H.E., Castro, J.M., Schipper, C.I., Gardner, J.E., Damby, D.E., McIntosh, I.M., Heap, M.J., Farquharson, J.I., Dingwell, D.B., Iacovino, K., Rooyakkers, S., **Kushnir, A.R.L.**, Carbillet, L., Kendrick, J.E., Lavallée, Y., Humphreys, M.C.S., Scheu, B., Weaver, J., Lamur, A., Brown, R.J., Ellis, B.S. (2023) From pyroclasts to lava: Silicic volcanism and explosive-effusive transitions. Presented at IAVCEI 2023 Scientific Assembly, 30 January-3 February 2023, Rotorua, New Zealand.

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15. Dezayes, C., Lerouge, C., **Kushnir, A.** (2019) Characterisation of the basement-sedimentary transition zone in the Saint Pierre Bois quarry (Vosges, France). Presented at Geological Association of Canada and the Mineralogical Association of Canada and the International Association of Hydrologists (GAC-MAC-IAH) General Assembly, 12-15 May 2019, Québec, Canada.
16. Ryan, A.G., Russell, J.K., Heap, M.J., Vona, A., Kolzenburg, S., **Kushnir, A.R.L.** (2019) The onset of bubble coalescence and melt fracturing in impermeable melt foams: implications for efficient outgassing of bubbly magmas. IUGG19-1583. Presented at 27th International Union of Geodesy and Geophysics (IUGG) General Assembly, 8-18 June 2019, Montreal, Canada.
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18. Farquharson, J., Baud, P., Wild, B., **Kushnir, A.R.L.** (2018) The potential for acid stimulation of a deep geothermal reservoir: an experimental investigation. EGU2018-7003. Presented at European Geosciences Union (EGU) 2018 General Assembly, 8-13 April 2018, Vienna, Austria.
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26. Komorowski, J-C., Jenkins, S., Charbonnier, S., Baxter, P.J., Gertisser, R., Cholik, N., Raditya, P., Surono, A,. Budi-Santoso, A., Preece, K., Devi S. Dayyudi, Jousset P., Martel, C., Arbaret, L., Burgisser, A., Erdmann, S., **Kushnir, A.,** Metaxian, J-P., Beauducel, F., A. Piquout, Lavigne, F., (2014). A review of timescales, processes, and impacts of explosive activity during the 2010 paroxysmal eruption of Merapi: evidence of a change in style? Presented at Cities on Volcanoes 8 (CoV8), 9-13 September 2014, Yogjakarta, Indonesia.
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Editorials

1. Chevrel, O., Wadsworth, F.B., Farquharson, J.I., **Kushnir, A.R.L.,** Heap, M.J., Williams, R., Delmelle, P., Kennedy, B. (2021): Publishing a Special Issue of Reports from the volcano observatories in Latin America. Editorial to "Special Issue on Volcano Observatories in Latin America" special issue in Volcanica. Volcanica 4(S1). DOI: https://doi.org/10.30909/vol.04.S1.ivi. *Citations: 7.*

2. **Kushnir, A.R.L.**, Loewer, M. (2020): From exploration to operation: Research developments in deep geothermal energy. Editorial to the “From exploration to operation” special issue in Geothermal Energy. Edited by Kushnir and Loewer. Geothermal Energy 8. DOI: https://doi.org/10.1186/s40517-020-00169-6. *Citations: 8.*

Technical and government reports

1. **Kushnir, A.**, Enkin, R.J., Welles, J., Kolzenburg, S. Porritt, L. Kopylova, M.G., Russell, J.K. (2013): Rock mechanical and geophysical properties of kimberlite: The roles of lithofacies and alteration. Unpublished technical report for Diavik Diamond Mines.
2. **Kushnir, A.**, Andrews, G., Russell, J.K., Enkin, R.J., Kennedy, L.A., Heap, M.J. and Quane, S. (2012): Rock physical-property measurements for the Nechako Basin oil and gas region, central British Columbia (parts of NTS 093B, C, E, F, G, K, L); in Geoscience BC Summary of Activities 2011, Geoscience BC, Report 2012-1, p. 125–150.
3. Andrews, G., Quane, S., Enkin, R.J., Russell, K., **Kushnir, A.**, Kennedy, L., Hayward, N. and Heap, M.J. (2011): Rock physical property measurements to aid geophysical surveys in the Nechako Basin oil and gas region, central British Columbia; Geoscience BC, Report 2011-10, 40 p.